SUBSTATION No. 35

MILLING CIRCUIT 2 FILTRATION/GOLDROOM

SUBSTATION EARTH BAR

415V FROM MCC 20-SW-62

20-SW-62

2000-F20-SLD-EL-10102

F66-SW-402

415V MOTOR CONTROL CENTRE

MCC EARTH BAR

NEUTRAL BAR

NOTE 1

415V 3NE~ 50Hz 800A 65kA for 1sec

1F-10	1F-13	1F-16	1F-19	1F-22	1R-22	1F-05	1R-10	1R-14	1R-18		
100A	100A	100A	100A	100A	100A	100A	20A	50A	50A		
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A		
/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >		
70A	70A	70A	70A	70A	70A	70A	70A	70A	70A		
						EOL	EOL	EOL	EOL		
F66-ER-401A	F66-ER-401B	F66-ER-411A	F66-ER-411B	F66-ER-412A	F66-ER-412B	F66-PS-481	F66-PW-460	F66-PW-461	F66-PW-462		
25kW	25kW	25kW	25kW	25kW	25kW	M 3~ 30kW	M 3~ 2.2kW	M 3~ 15kW	M 3~ 15kW		
GRAVITY ELECTROWINNING CELL RECTIFIER 1 2000-F66-SCM-EL-10201	GRAVITY ELECTROWINNING CELL RECTIFIER 2 2000-F66-SCM-EL-10202	CIL 4 ELECTROWINNING CELL 1 RECTIFIER 1 2000-F66-SCM-EL-10206	CIL 4 ELECTROWINNING CELL 1 RECTIFIER 2 2000-F66-SCM-EL-10207	CIL 4 ELECTROWINNING CELL 2 RECTIFIER 1 2000-F66-SCM-EL-10210	CIL 4 ELECTROWINNING CELL 2 RECTIFIER 2 2000-F66-SCM-EL-10211	ELECTROWINNING TANK AREA SUMP PUMP 2000-F66-SCM-EL-10209	GRAVITY ELECTROWINNING PUMP 2000-F66-SCM-EL-10208	CIL 4 ELECTROWINNING PUMP 1 2000-F66-SCM-EL-10203	CIL 4 ELECTROWINNING PUMP 2 2000-F66-SCM-EL-10204		

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM 20-SW-62 INCOMER. SIGNAGE SHALL BE PLACED ON F66-SW-402 INCOMING SUPPLY SECTION READING "M.E.N. LINK LOCATED IN 20-SW-62".
2. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
3. ALL DRIVE I/O VIA COMMS NETWORK TO DCS.
4. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
5. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC
- EOL — ELECTRONIC OVERLOAD UNIT
- / > — EARTH LEAKAGE RELAY
- FIELD-MOUNTED DECONTACTOR
- CIRCUIT BREAKER (NOTE 5)
  - xxxA — ST TRIP SET POINT
  - xxxA — CB SIZE

NORTHERN STAR RESOURCES LTD

Approved	x
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

2000-F66-SLD-EL-10101_Rev1

Milling CCT 2 filtration/goldroom substation no. 35
MCC F66-SW-402 single line diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F66 - ELECTROWINNING & GOLDROOM\2000-F66-SLD-EL-10101.DWG 13NOV25 12:39PM HOTSEAT04